121. A clock spring is heated to redness and then plunged into cold water. This treatment will cause it to become

(a) Soft and ductile
(b) More springy than before
(c) Strongly magnetic
(d) Hard and brittle

Answer: D — (d) Hard and brittle

Chemical reason: Rapidly cooling red-hot spring steel is quenching. It forms a hard, brittle martensitic structure; tempering would be required afterward to restore springiness.

222. Mark the steel in which carbon % is highest

(a) Mild steel
(b) Hard steel
(c) Alloy steel
(d) None of these

Answer: B — (b) Hard steel

Chemical reason: Hard steel contains more carbon than mild steel; higher carbon generally increases hardness and strength but lowers ductility.

223. Mark the variety of iron which has highest melting point

(a) Pig iron
(b) Cast iron
(c) Wrought iron
(d) Steel

Answer: C — (c) Wrought iron

Chemical reason: Wrought iron is the purest commercial iron among these forms and therefore has the highest melting point; carbon and impurities lower the melting range of pig/cast iron and steel.

224. Bessemer converter is used in the manufacture of

(a) Pig iron
(b) Steel
(c) Wrought iron
(d) Cast iron

Answer: B — (b) Steel

Chemical reason: A Bessemer converter is used to convert molten pig iron into steel by blowing air through it, oxidizing excess carbon and other impurities.

225. Steel contains

(a) Fe+C+Mn
(b) Fe+C+Al
(c) Fe+Mn
(d) Fe+Mn+Cr

Answer: A — (a) Fe+C+Mn

Chemical reason: This is determined by the standard composition or heat-treatment behaviour of the metal/alloy. The description in Fe+C+Mn matches the relevant metallurgical process or alloy property.

226. Steel differs from pig iron and wrought iron in that it contains

(a) No carbon at all

(b) Less carbon than either

(c) More carbon than either

(d) An amount of carbon intermediate between two

Answer: D — (d) An amount of carbon intermediate between two

Chemical reason: Steel has a carbon content intermediate between wrought iron (very low carbon) and pig/cast iron (high carbon). That intermediate composition gives steel its characteristic balance of strength and ductility.

227. Finely divided iron combines with CO to give

(a) $Fe(CO)_5$

(b) $Fe_2(CO)_9$

(c) $Fe_3(CO)_{12}$

(d) $Fe(CO)_6$

Answer: A — (a) $Fe(CO)_5$

Chemical reason: This is determined by the standard composition or heat-treatment behaviour of the metal/alloy. The description in $Fe(CO)_5$ matches the relevant metallurgical process or alloy property.

228. Mohr's salt is

(a) $FeSO_4\cdot 7H_2O$

(b) $Fe(NH_4)SO_4\cdot 6H_2O$

(c) $(NH_4)_2SO_4.FeSO_4\cdot 6H_2O$

(d) $[Fe(NH_4)_2](SO_4)_2\cdot 6H_2O$

Answer: C — (c) $(NH_4)_2SO_4.FeSO_4\cdot 6H_2O$

Chemical reason: The correct choice is $(NH_4)_2SO_4.FeSO_4\cdot 6H_2O$. It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.

229. Mohr's salt is

(a) Normal salt

(b) Acid salt

(c) Basic salt

(d) Double salt

Answer: D — (d) Double salt

Chemical reason: Mohr’s salt is a double salt because it crystallizes from ferrous sulfate and ammonium sulfate in a definite ratio but dissociates in water into the constituent ions.

230. An example of double salt is

(a) Bleaching powder

(b) $K_4[Fe(CN)_6]$

(c) Hypo

(d) Potash alum

Answer: D — (d) Potash alum

Chemical reason: Potash alum, $KAl(SO_4)_2\cdot 12H_2O$, is a classic double salt: in solution it gives K^+, Al^{3+}, and SO_4^{2-} ions rather than retaining a complex ion.

231. The passivity of iron in concentrated nitric acid is due to

(a) Ferric nitrate coating on the metal
(b) Ammonium nitrate coating on the metal
(c) A thin oxide layer coating on the metal
(d) A hydride coating on the metal

Answer: C — (c) A thin oxide layer coating on the metal

Chemical reason: Concentrated HNO_3 passivates iron by producing a thin, adherent oxide film on the surface. This film blocks further reaction with the acid.

232. The action of steam on heated iron is represented as

(a) $3Fe+_4H_2O \rightarrow Fe_3O_4+_4H_2$
(b) $2Fe+_3H_2O \rightarrow Fe_2O_3+_3H_2$
(c) $Fe+H_2O \rightarrow FeO+H_2$
(d) $2Fe+H_2O+O_2 \rightarrow Fe_2O_3+H_2$

Answer: A — (a) $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$

Chemical reason: This is determined by the standard composition or heat-treatment behaviour of the metal/alloy. The description in $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$ matches the relevant metallurgical process or alloy property.

233. Which metal is used to make alloy steel for armour plates, safes and helmets

(a) Al
(b) Mn
(c) Cr
(d) Pb

Answer: B — (b) Mn

Chemical reason: Manganese steel is very hard and impact-resistant, making it useful for armour plates, safes, helmets, rails, and similar heavy-duty applications.

234. Rusting on iron needs

(a) Dry air
(b) Air and water
(c) Distilled water and carbon dioxide
(d) Oxygen and carbon dioxide

Answer: B — (b) Air and water

Chemical reason: Rusting is an electrochemical process that requires both water and oxygen. Water provides the electrolyte, while dissolved oxygen participates in the cathodic reaction.

235. Iron when treated with concentrated nitric acid

(a) Readily reacts
(b) Slowly reacts
(c) Becomes passive
(d) Gives ferrous nitrate

Answer: C — (c) Becomes passive

Chemical reason: Concentrated nitric acid makes iron passive because an oxide film forms on the surface and prevents continued attack.

236. An alloy which does not contain copper is

(a) Solder

(b) Bronze

(c) Brass

(d) Bell metal

Answer: A — (a) Solder

Chemical reason: Solder is mainly an alloy of lead and tin and therefore contains no copper. Bronze, brass, and bell metal are copper-based alloys.

237. Which one of the following statements shows the correct percentage of carbon in steel, pig iron and wrought iron

(a) Steel containing less than 0.15% carbon; wrought iron 0.15 to 2.0% carbon; and pig iron over 2% carbon

(b) Pig iron less than 0.15% carbon; wrought iron 0.15 to 2.0% carbon; and steel over 2% carbon

(c) Wrought iron less than 0.15% carbon; steel 0.15 to 2.0% carbon; and pig iron over 2% carbon

(d) Wrought iron less than 0.15% carbon; pig iron 0.15 to 2.0% carbon; and steel over 2.0% carbon

Answer: C — (c) Wrought iron less than 0.15% carbon; steel 0.15 to 2.0% carbon; and pig iron over 2% carbon

Chemical reason: The standard carbon-content order is wrought iron < steel < pig iron. Thus wrought iron has less than about 0.15% C, steel roughly 0.15–2%, and pig iron above 2%.

238. In the Bessemer and open–hearth process for the manufacture of steel, which one of the following is used for the removal of carbon in part or whole Bessemer Open– hearth

(a) Ferric oxide Air

(b) Air Ferric oxide

(c) Oxygen Scrap iron

(d) Air Scrap iron

Answer: D — (d) Air Scrap iron

Chemical reason: This is determined by the standard composition or heat-treatment behaviour of the metal/alloy. The description in Air Scrap iron matches the relevant metallurgical process or alloy property.

239. About the basic open hearth process, which statement is wrong

(a) Limestone is added to the charge

(b) Phosphorus impurity cannot be removed by this process

(c) Carbon content of the steel can be uniformly controlled over a series of batches

(d) Iron scrap can be utilized

Answer: A — (a) Limestone is added to the charge

Chemical reason: In the basic open-hearth process, phosphorus can be removed because a basic lining/flux such as lime converts phosphorus oxides into slag. Therefore the statement saying phosphorus cannot be removed is wrong.

240. Which of the following statements is wrong

(a) Heating to a high temperature and then cooling suddenly, e.g. by dipping in water, makes steel hard and brittle

(b) Steel can be softened by heating it to a high temperature for a prolonged time and then cooling slowly. This is called quenching.

(c) Tempering of hardened steel is done by heating it to just below red heat at controlled temperature and duration

(d) Phosphorus impurity makes steel 'cold short'

Answer: D — (d) Phosphorus impurity makes steel 'cold short'

Chemical reason: Slow cooling after prolonged heating is annealing, not quenching. Quenching means rapid cooling, usually in water or oil, so the statement using "quenching" for slow cooling is incorrect.